SATGURU STUDY CENTRE

Chennai - 81

# TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

SATGURU STUDY CENTRE
NEVER STOP LEARNING.NEVER STOP GROWING...
17/8, Poondi Thangammal St, Nagoor Garden, Tondiarpet, Chennai, Tamil Nadu 600081
An MSME-Certified Learning Centre

CLASS : 11

Number

# UNIT TEST - 3,NOVEMBER - 2024

## PHYSICS

Time Allowed : 1.30 Hours] [Max. Marks : 35

### PART - I

**I. Answer all the questions** 5x1=5

1. Which of the following is not a scalar?

   a) viscosity b) surfacetension c) pressure d) stress

2. The graph between volume and temperature in Charle's law is

   a) an ellipse b) a circle c) a straight line d) a parabola

3. In an Isochoric Process, we have

   a) $W = 0$ b) $Q=0$ c) $\Delta U=0$ d) $\Delta T = 0$

4. A distant star emits radiation with maximum intensity at 350 nm. The temperature of the Star is

   a) 8280 K b) 5000 K c) 7260 K d) 9044 K

5. If a wire is stretched to double of its original length, then the strain in the wire is

   a) 1 b) 2 c) 3 d) 4

### PART - II

**II. Answer any three questions. Question no.10 is Compulsory.** 3x2=6

6. What is Reynold's Number?

7. Distinguish between Cohesive Force and Adhesive Force.

8. Which one of these is more elastic steel or rubber? Why?

9. State Stefan - Boltzman Law.

10. A refrigerator has COP of 3. How much work must be supplied to the refrigerator in order to remove 200 J of heat from its interion?

## PART - III

**III. Answer any three of the following. Q.No.15 is Compulsory.** **3x3=9**

11. Write the conditions for Reversible Process.

12. Draw the PV diagram for a) Isothermal process and b) Isochoric Process.

13. Discuss various modes of Heat Transfer.

14. Derive the expression for the terminal velocity of a sphere moving in a high viscous fluid using Stokes force.

15. Let $2.4 \times 10^{-4}$ j of work is done to increase the area of a film of Soap bubble from 50 $cm^2$ to 100 $cm^2$. Calculate the value of surface tension of soap solution.

## PART - IV

**IV. Answer the following in detail:** **3x5=15**

16. a) State and Prove Bernoullis theorem for a flow of incompressible, non viscous and streamlined flow of fluid.

    **(OR)**

    b) Obtain an expression for the surface tension of a liquid by capillary rise method.

17. a) Explain in detail Newton's law of cooling.

    **(OR)**

    b) Derive the Expression for Carnot engine efficiency.

18. a) Derive Poiseuille's formula for the volume of a liquid flowing per second through a Pipe under stream lined flow.

    **(OR)**

    b) Explain in detail the thermal expansion of Solid, Liquid and Gas.